CHE 565 -- Homework 5

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Problem 1

Closed-loop Response to Step Disturbance

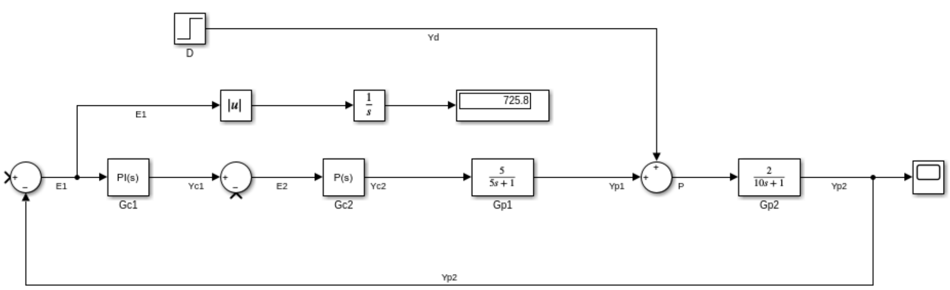
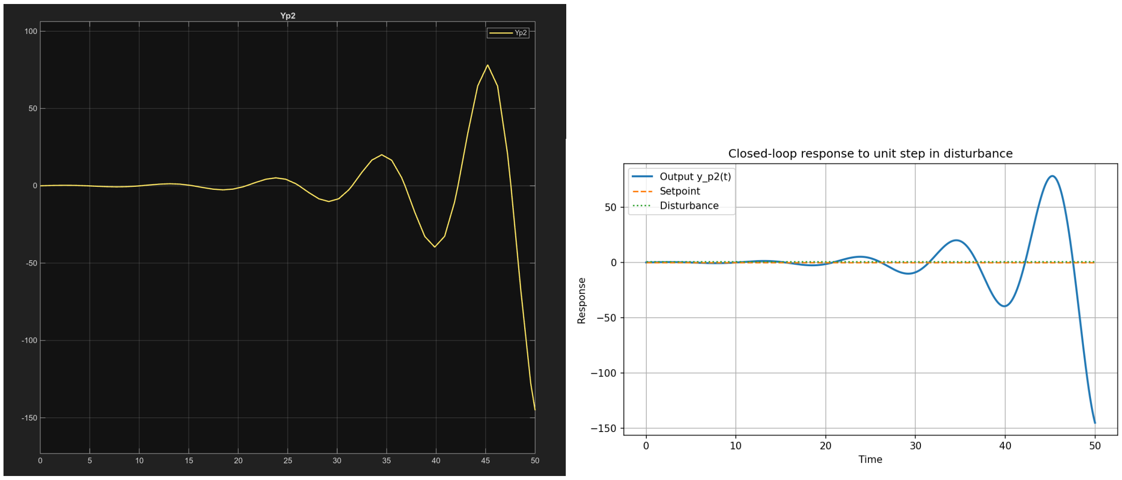


Figure 1: Block diagram of the closed-loop system without cascade control and with a unit step disturbance from Simulink.

The Simulink block diagram is shown in Figure 1.

Case 1: Step disturbance with no setpoint change gave and step in disturbance with no setpoint change gaveIAE = 725.3696 in Python and IAE = 725.8109 in Simulink.



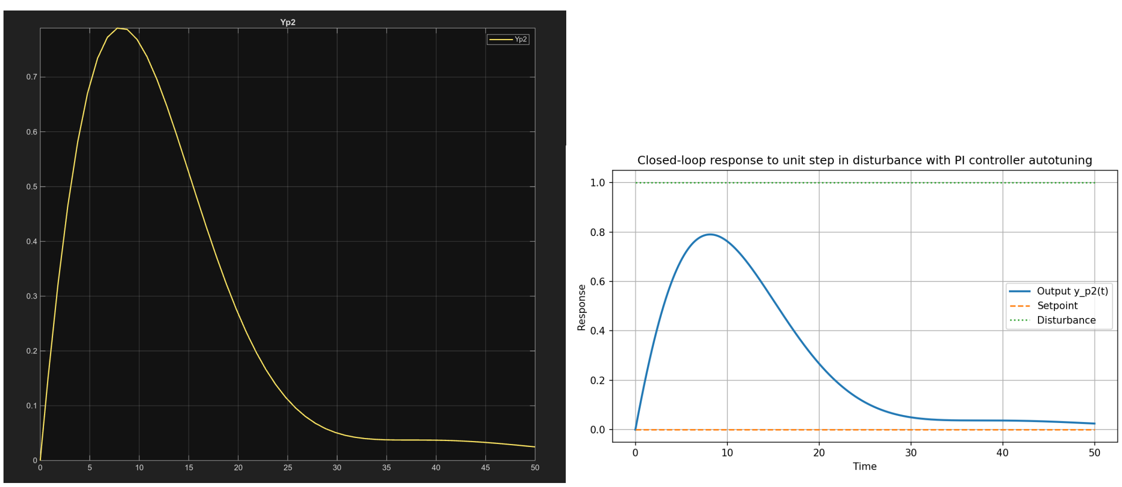
(a) Simulink (b) Python

Figure 2: Closed-loop response to no step change in setpoint and a unit step change in disturbance with no cascade control and no autotuning.

The closed-loop disturbance response and no cascade loop is shown in Figure 2.

The PI controller was then tuned using the autotuning feature in Simulink, which gave P = 0.1837 and I = 0.0816.

Case 2: Step disturbance with no setpoint change and PI autotuning gave IAE = 13.0462 in Python and IAE = 13.0463 in Simulink.

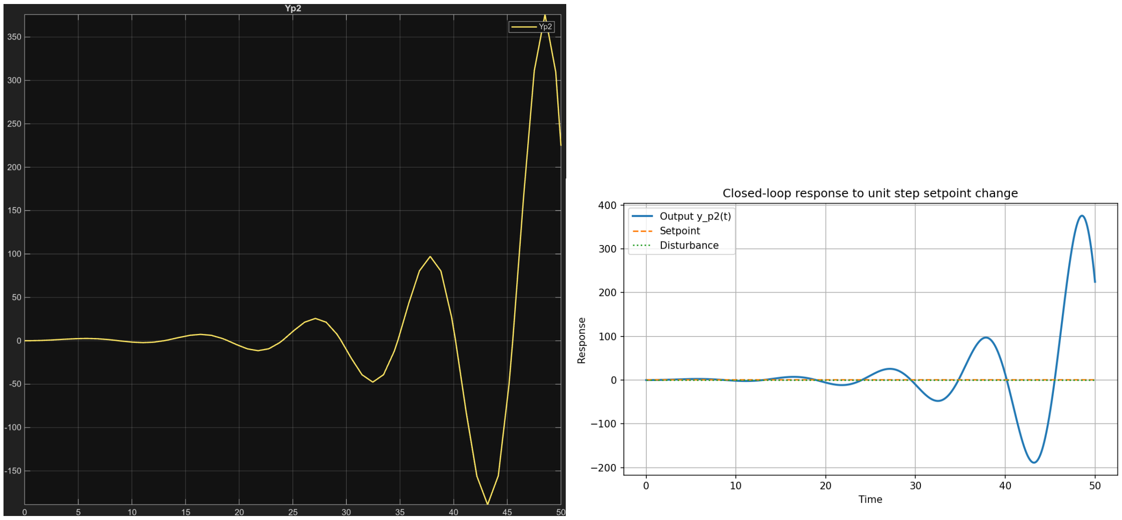


(a) Simulink (b) Python

Figure 3: Closed-loop response to no step change in setpoint and a unit step change in disturbance with autotuning.

The response to a unit step change in disturbance with no cascade control and PI autotuning is shown in Figure 3.

Closed-loop Response to Step Setpoint Change

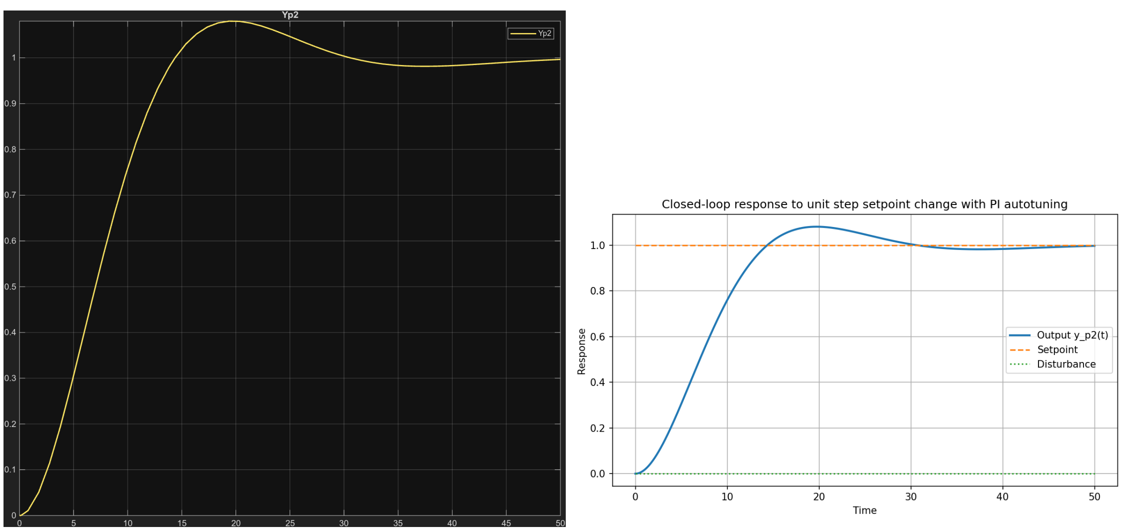


(a) Simulink (b) Python

Figure 4: Closed-loop response to step change in setpoint and no step disturbance.

The response to the step change in setpoint with no step disturbance is shown in Figure 4.

Case 3: Step setpoint change with no disturbance change gave IAE = 2449.4065 in Python and IAE = 2450.8205 in Simulink.



(a) Simulink (b) Python

Figure 5: Closed-loop response to step change in setpoint and no step disturbance with PI autotuning.

The response to a unit step change in setpoint with no disturbance change using PI autotuning is shown in Figure 5.

Case 4: Step setpoint change with no disturbance change and PI autotuning gave IAE = 8.1939 in Python and IAE = 2450.8205 in Simulink.